ALGEBRA 2 · PROBABILITY & STATISTICS

Probability Simulation Lab

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Probability Simulation Lab** on Mr. K's Classroom: Algebra 2 → Probability & Statistics → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. If you flip a coin 10 times, will you always get exactly 5 heads?

   No. Small samples vary a lot.

2. Which sum of two dice do you think comes up most often?

   7

2 Explore use the tool and record what you see

Coin flips

Keep adding flips and record the share of heads.

Total flips	Heads	Experimental P(heads)	Difference from 0.5
10	**answers vary**	**answers vary**	**answers vary**
100	**answers vary**	**answers vary**	**usually smaller**
1,000	**answers vary**	**answers vary**	**usually smaller still**
3,000	**answers vary**	**close to 0.5**	**usually under 0.02**

Sum of two dice: theoretical probability

Compare with your experimental bars after 500+ rolls.

Sum	2	4	6	7	9	12
Ways (out of 36)	**1**	**3**	**5**	**6**	**4**	**1**
Probability	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{4}{36}$	$\frac{1}{36}$

Normal curve: μ = 70, σ = 8

Shade within	Interval	Percent
1σ	**62 to 78**	**about 68%**
2σ	**54 to 86**	**about 95%**
3σ	**46 to 94**	**about 99.7%**

3 Write the rule fill in each blank from what you observed

1. Experimental probability = $\frac{\text{times it happened}}{\text{number of trials}}$. Theoretical probability = $\frac{\text{favorable outcomes}}{\text{possible outcomes}}$.

2. Law of large numbers: as the number of trials grows, the experimental probability gets **closer** to the theoretical probability.

3. Empirical rule: about **68** % of data lies within 1σ, **95** % within 2σ and **99.7** % within 3σ.

4. Increasing σ makes the normal curve **wider** and **shorter**.

4 Check no tool: show your work

1. What is P(sum = 8) for two dice?

   $\frac{5}{36}$

2. Scores have μ = 80 and σ = 5. What percent fall between 75 and 85?

   about 68%

3. Using the same scores, what percent are above 90?

   about 2.5%

4. You roll one die 600 times. About how many 4s do you expect?

   100

5 Reflect

Why did 10 flips sometimes give a share far from 0.5 while 3,000 flips did not?

With few trials, chance has a big effect; with many trials the random ups and downs average out.